

AP CSP | Lesson 06: Compression

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C is correct: $18 \times 8 = 144$ bits. A adds 8 rather than multiplying by the number of bits per byte. B uses four bits per byte. D uses ten bits per byte. One byte contains eight bits.

Q2. Answer: A

A is correct: copy receives 80 before original changes to 100. B incorrectly treats copy as a live link to original. C subtracts the two values, but the program contains no subtraction. D adds them, but the program contains no addition. A variable's current value follows its actual assignments, not a guess from its name.

Q3. Answer: D

D is correct: the guarantee of exact reconstruction defines lossless compression. A confuses a different representation with information loss. B overlooks that a more efficient representation may retain the same information in fewer bits. C extends one file's size reduction to all possible inputs, which is not justified.

Q4. Answer: B

B is correct: the sequence has two A's, then three B's, then two C's. A exchanges the first two run counts. C preserves the counts but moves the C run before the B run. D uses letter-count order rather than the specified count-letter format, so it is not a valid encoding under these rules.

Q5. Answer: B and D

B is correct because different representations contain different patterns, and algorithms exploit different kinds of redundancy. D is correct because the changed representation can preserve everything needed for exact reconstruction. A is false: the toy RLE expands alternating symbols. C attributes the lossless guarantee to a lossy method. Award one mark only for B and D together and no other choices.

Q6. Answer: A

A is correct: $(240 - 180) / 240 \times 100\% = 25\%$. B divides the 60-byte saving by the encoded size 180 rather than the original 240. C reports the numeric byte saving as though it were a percentage. D is the encoded size as a percentage of the original, $180 / 240 = 75\%$, not the percentage saved.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) 4A5B2C. (b) Raw: 11 bytes. Encoded: 6 bytes. (c) The decoder reads the count-letter pairs and produces four A's, then five B's, then two C's, preserving every original character in order.

1 mark: Writes 4A5B2C using the specified count-letter order.

1 mark: Gives the raw size as 11 bytes.

1 mark: Gives the encoded size as 6 bytes, with three two-byte pairs or equivalent reasoning.

1 mark: Explains that expanding each count-letter pair in order restores the exact original sequence.

Q8. 4 marks

Answer: (a) CCCABB. (b) Six reconstructed letters; six encoded bytes. (c) 2A2B decodes to AABB, so it changes the original order. A correct encoding is 1A1B1A1B.

1 mark: Decodes 3C1A2B as CCCABB.

1 mark: Gives both six reconstructed letters and six encoded bytes.

1 mark: Explains that 2A2B decodes to AABB and therefore changes the order of ABAB.

1 mark: Gives the correct encoding 1A1B1A1B.

Q9. 4 marks

Answer: (a) 1A1B1C1A1B1C. (b) Raw: 6 bytes; encoded: 12 bytes. (c) $(12 - 6) / 6 \times 100\% = 100\%$ increase. (d) This RLE exploits consecutive identical symbols, which this input lacks. A different algorithm might exploit its repeated ABC pattern.

1 mark: Writes 1A1B1C1A1B1C.

1 mark: Gives both 6 raw bytes and 12 encoded bytes.

1 mark: Gives a 100% increase using the original 6 bytes as denominator.

1 mark: Explains that another algorithm may exploit a different pattern, such as repeated ABC, while this RLE only exploits consecutive identical letters.

Q10. 4 marks

Answer: (a) 14 bytes. (b) 6 bytes saved. (c) $(20 - 14) / 20 \times 100\% = 30\%$. (d) The header is part of the stored encoded file and must be counted; using only the payload would understate its size and overstate the saving.

1 mark: Adds the 10-byte payload and 4-byte header to obtain 14 bytes.

1 mark: Subtracts from 20 to obtain 6 bytes saved.

1 mark: Calculates 30% saved with original size 20 as denominator.

1 mark: Explains that the required header occupies storage and excluding it understates the encoded size or overstates the saving.

Q11. 4 marks

Answer: (a) Both store 3. (b) Both reconstruct as 30. (c) Lossy: distinct inputs 31 and 38 produce the same stored digit, so that digit cannot identify which exact original value was present. (d) No. Correct recovery for one input does not guarantee exact recovery for every allowed input.

1 mark: States that both 31 and 38 store the digit 3.

1 mark: States that both reconstruct as 30.

1 mark: Identifies the method as lossy and explains that the stored data cannot distinguish or exactly recover the two different originals.

1 mark: Says no and explains that exact recovery for 30 does not guarantee exact recovery for all allowed inputs.

Q12. 4 marks

Answer: (a) Choose L because it guarantees the exact recovery required by the archive. (b) Choose P because 160 KB is under 200 KB and its approximation has been approved for the preview purpose.

1 mark: Chooses option L for the archive.

1 mark: Justifies L with its guarantee of exact reconstruction, matching the archive requirement.

1 mark: Chooses option P for the preview system.

1 mark: Justifies P by citing both its 160 KB size below 200 KB and the acceptability of the approved approximation.